4.3 Related Rates

Name _____ Period ____ Date _____

AP Calculus 4.3 (30-minute plan) · Student sheet

I can…

- write an equation that relates the changing quantities.
- differentiate an equation with respect to time t.
- solve related rate problems with circles, spheres, ladders, and cones.

Vocabulary · fill in the missing word

Related rates — rates of change of _____ that are related by an equation

Rate with respect to time — $\frac{dx}{dt}$, how fast x _____ per unit of time

Part 1 · Notes and examples (fill in each note, then work the example)

Step 1: draw a picture, name the variables, and write the given rates and the _____ rate.

Step 2: write an equation that relates the _____.

Example 1

The radius of a circle increases at 2 cm/s. How fast is the area increasing when r = 10 cm?

Step 3: differentiate both sides with respect to _____.

Step 4: substitute the values at the instant only _____ differentiating.

Example 2

A 10-ft ladder leans against a wall. The bottom slides away at 1 ft/s. How fast is the top changing when the bottom is 6 ft from the wall?

Example 3

Water pours into an inverted cone (radius 2 m, height 4 m) at 3 m^3/min. How fast is the water rising when it is 2 m deep?

Part 2 · Partner practice

Partner practice

(AP® Practice) A spherical balloon is inflated at $32\pi \ \frac{\text{cm}^3}{\text{s}}$. How fast is the radius increasing when r = 2 cm?

(A) 2 cm/s (B) $\frac{1}{2}$ cm/s (C) 8 cm/s (D) 4 cm/s